SIMATS ENGINEERING
ASSESSMENT TOOLS

COURSE NAME: **Theory of Computation** **COURSE CODE:** **CSA13**

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 40%

CO4 - AT 3

QUESTIONS: 5
Duration : 1 Hour

Total Marks:25

CLASS TEST

Code of Conduct:

I Om Sai Reddy T / 192372252 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

SIMATS ENGINEERING
ASSESSMENT TOOLS

1. Design a PDA to accept strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer.
i) $L = \{a^n b^n\}$

2. Convert the given context-free grammar (CFG) into a Pushdown Automaton (PDA) that recognizes the same language.

$E \rightarrow E+E \mid E*E \mid (E) \mid I$

$I \rightarrow Ia \mid Ib \mid I0 \mid I1 \mid a \mid b$

3. Design a TM to accepts the strings with 'n' occurrences of 'a' followed by 'n' occurrences of 'b', where 'n' is a non-negative integer
i) $L = \{0^n 1^n \mid n \geq 1\}$

4. Design a Pushdown Automata for the language representing a's followed by b's and the number of b's must be twice than that of a's.

$$L = \{a^n b^{2n} \mid n \geq 1\}$$

5. Construct a Turing Machine to recognize the language.Write the logic used for the design and the formal definition

$$L = \{a^n b^n c^n \mid n \geq 1\}$$

COY-AT ③

Given $L = \{a^n b^n\}$

To design a PDA to accept the strings with "n" occurence of a and followed by 'n' occurences of 'b'.

Where n is non-negative integer.

Idea:-

→ For Every 'a' push one Symbol A onto the stack.

⇒ for Every 'b' pop one 'A'.

⇒ Accept when all the input is read and only the bottom stack symbol remains.

PDA definitions:-

$$\boxed{M = \{Q, \Sigma, \Gamma, \delta, s, F, q_0\}}$$

where $Q = q_0, q_1, q_2$ [Set of Finite states].

$\Sigma = \{a, b\}$

$\Gamma = A; Z_0$

$S = q_0$ → Start State

$F = q_F$ → Final State.

$q_0 = Z_0$ → Initial Stack Symbol.

Transitions:-

δ = Transitions

$\delta(q_0, a, Z_0) = (q_0, AZ_0)$

$\delta(q_0, a, A) = (q_0, AA)$

$\delta(q_0, b, A) = (q_1, \varepsilon)$

$\delta(q_0, b, A) = \{q_1, \varepsilon\}$

$$\delta(q_1, \varepsilon, Z_0) = (q_F, Z_0).$$

Example :-

For input aaabbb :-

$$q_0 \xrightarrow{aaa} \text{Push AAA} \xrightarrow{bbb} \text{Pop AAA} \longrightarrow q_F$$

Input aaabbb

Read aaa → Push AAA.

Stack becomes empty → Accepted.

② Convert CFG into PDA

Given:-

CFG

$$E \rightarrow E+E \mid E^{*}E \mid (E) \mid I$$

$$I = Ia \mid Ib \mid I0 \mid I1 \mid a \mid b.$$

Method:-

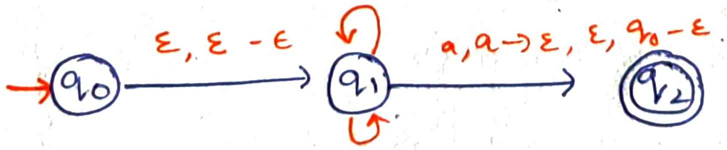
→ Put start symbol E into stack.

→ Replace variables using CFG productions.

→ Match terminals with input symbols and pop them.

→ When stack becomes empty accepted them.

PDA for CFG :-



Construction Method :-

Step ① :- Initialize the stack.

Push the Start Symbol E.

$$\delta(q_0, \varepsilon, z_0) = (q_r, E\text{-}z_0)$$

Step ② :- Replace variables using productions

For E :- $\delta(q_1, \varepsilon, E) = \{q_1, E+E\}$

$\delta(q_1, \varepsilon, E) = \{q_r, E*E\}$

$\delta(q_1, \varepsilon, E) = \{q_r, (E)\}$

$\delta(q_r, \varepsilon, E) = \{q, I\}$

For I :- $\delta(q, \varepsilon, I) = (q_1, Ia)$

$\delta(q, \varepsilon, I) = \{q_r, Ib\}$

$\delta(q, \varepsilon, I) = \{q_r, I0\}$

$\delta(q, \varepsilon, I) = \{q_1, I1\}$.

Step ③ :- Match ~~terminals~~

$\delta(q_r, a, a) \rightarrow \varepsilon$

$\delta(q_1, b, b) \rightarrow \varepsilon$

$\delta(q, 0, 0) \rightarrow \varepsilon$

$\delta(q_1, 1, 1) \rightarrow \varepsilon$

$\delta(q, *, *) \rightarrow \varepsilon$

$\delta(q, c, c) \rightarrow \varepsilon$

$\delta((q_1)) = \varepsilon$

Step ④ :- Accept

$\delta(q, \varepsilon, z_0) = (q_f, z_0)$.

The PDA simulates the CFG by replacing variables using the productions and matching terminals with the input.

(3)

Turning Machine For

$$L = \{0^n, 1^n) \mid n \geq 1\}$$

⇒ Find the leftmost unmarked 0 and replace it with x.

⇒ More light and find first unmarked 1; replace it with

⇒ Return to left.

⇒ Repeat until all 0's and 1's are marked.

⇒ Accept if the no.of 0's and 1's are equal.

$$\boxed{M = \{Q, \Sigma, \delta, \Gamma, q_0, S, F\}}$$

where $Q = \{q_0, q_1, q_2, q_3, q_{accept}, q_{reject}\}$

$\Sigma = \{0, 1\}$

$\Gamma = \{0, 1, x, y, \delta\}$

$F = \{q_{accept}\}$

Transitions:-

$\delta(q_0, 0) = (q_1, x, R)$

$\delta(q_0, x) = (q_0, x, R)$

$\delta(q_1, 0) = (q_0, R)$

$\delta(q_1, x) = (q, y, R)$

$\delta(q, 1) = (q_2, y, L)$

Return left:-

$\delta(q_2, 0) = (q_2, 0, 1)$

$\delta(q_2, 7) = (q_2, L, L)$

$\delta(q_2, x) = (q_0 x, R)$

Accept: when all symbols are marked.

$\delta(q_0, y) = (q_3, y, R)$

$\delta(q_3, y) = (q_2, y, R)$

Ex:- 0011

0011 → X0Y1 → xxyy

∴ The String is accepted.

Margin notes: x; every; for; for

PDA for $L = \{a^n b^{2n} \mid n \geq 1\}$

→ for every a, push two stack symbols.

→ for every b, pop one stack symbol.

→ Therefore if 'n' occurance of 'a' the stack contains 2^n symbols and exactly 2n bs are required to empty it.

$$\boxed{M = \{Q, \Sigma, \delta, \Gamma, q_0, Z_0, F\}}$$

where $Q_0 = \{q_0, q_1, q_2\}$

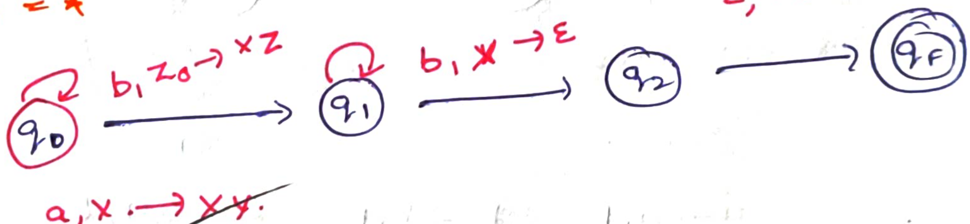
$\Sigma = \{a, b\}$

$\Gamma = A, Z_0$

$F = q_F$

PDA diagram:-

=*=*=*=*



Transitions:-

Current	Input	Stack	Actions	Next
q_0	a	Z_0	AAZ_0	q_0
q_0	a	A	AAA	q_0
q_0	b	A	ε	q_1
q_1	b	A	ε	q_1
q_1	ε	Z_0	ε	F

Example:-

a^2b^4 = aabbbb

Stack Operations

a → AA

a → AAA

b → POP A

b → POP A

b → POP A

b → POP A.

stack → Z_0.

∴ Accepted.

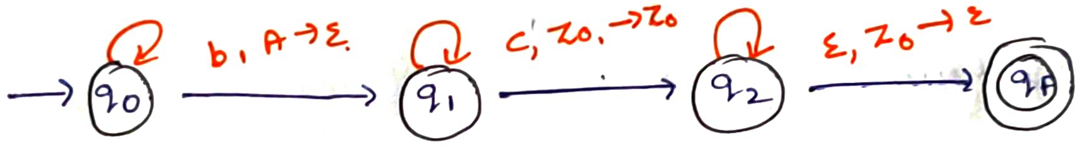
⑤ TM for $L = \{a^n b^n c^n \mid n \geq 1\}$

For every a:

- ⇒ Change a to x.
- ⇒ Find one b and change it to y.
- ⇒ Find one c and change it to z.
- ⇒ Return to left.
- ⇒ Repeat.
- → If all a, b and c are matched ⇒ accept.

$$\boxed{M = \{Q, \Sigma, \Gamma, \delta, q_0, z_0, F\}}$$

TM Diagram:-



Important transitions:-

q0	q1	q2	qF	
a → x, R	a → a, R	b → b, R	a → a, L	z → z, L
x → x, R	x → x, R	y → y, R	b → b, L	B → B, R
	b → y, R	ε → z, L	c → c, L	
			x → x, L	
			y → y, L	

Example :-

Input aaa bbb ccc.

↓

$\boxed{\text{x a a b y b b z c c}}$ (First round)

↓

$\boxed{\text{x x a x x b z z c}}$ (Second round)

↓

$\boxed{\text{x x x y y y z z z}}$ (Third round).

Accepted.

∴ The string is accepted.

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...RICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

86/100